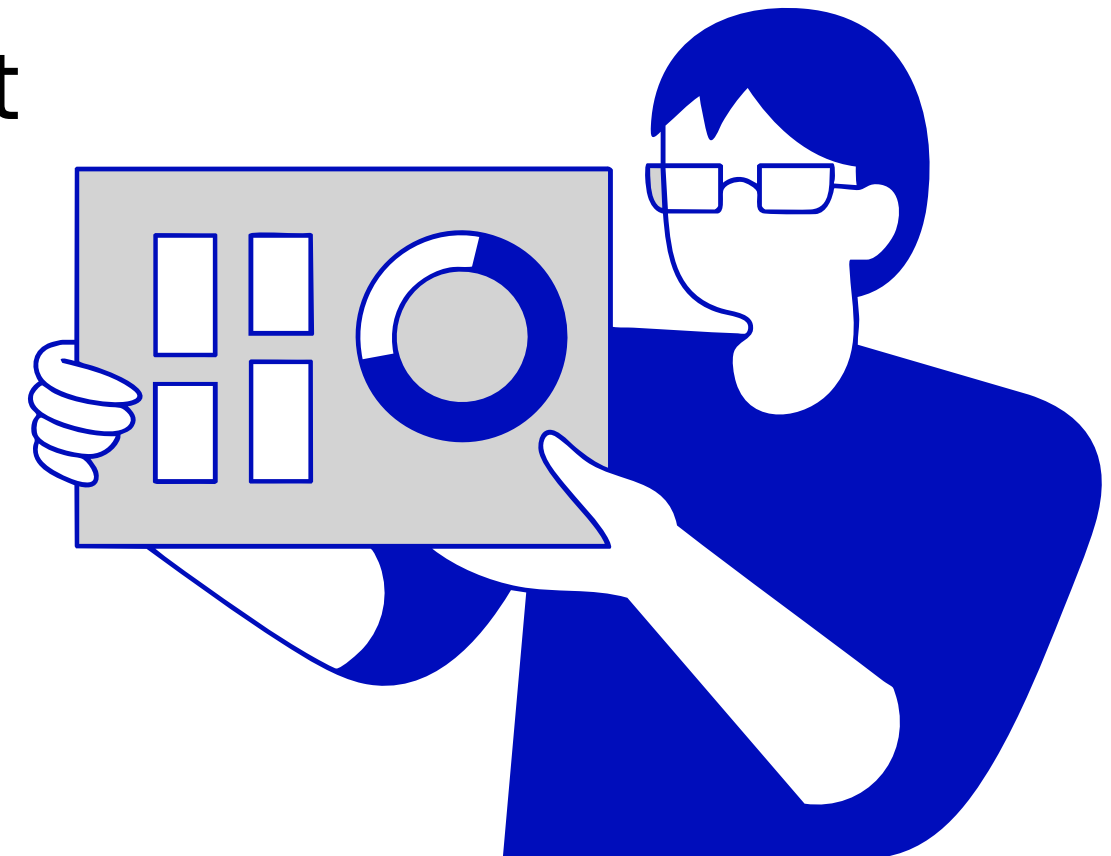


DataBase Management System



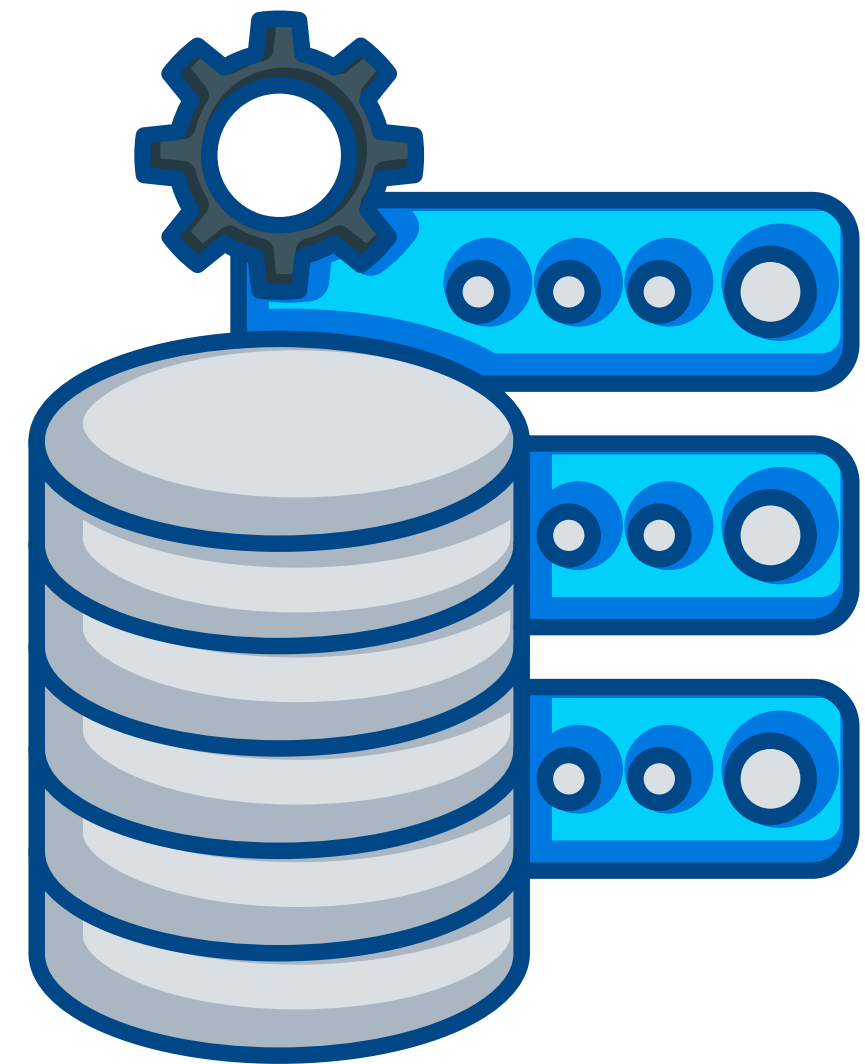
Data

- Data refers to raw facts, figures, or symbols that represent information.
- It can be numbers, text, images, audio, or video that have no meaning on their own but can be processed to extract **useful insights**.



Database

- A database is an organized **collection of related data** stored in a structured way.
- It allows for easy access, management, updating, and retrieval of data.



Relational vs Non-Relational

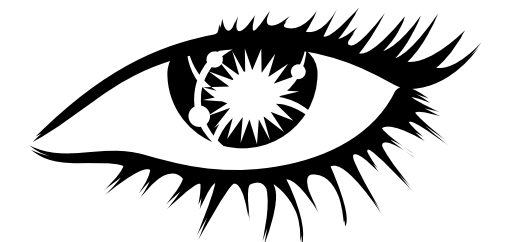
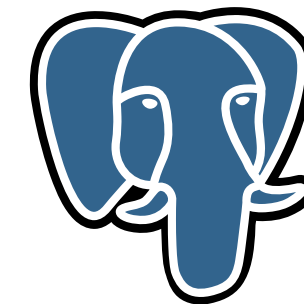
Relational Databases (RDBMS)

- Data is stored in tables (rows and columns)
- Uses SQL to manage data
- Maintains relationships using keys
- Examples: MySQL, PostgreSQL, Oracle



Non-Relational Databases (NoSQL)

- Data stored in flexible formats (documents, key-value, graphs, etc.)
- Ideal for unstructured or semi-structured data
- Scales easily for big data and real-time applications
- Examples: MongoDB, Cassandra, Redis



ORACLE



Structured Query Language (SQL)

SQL (Structured Query Language) is a standard programming language used to manage and manipulate **relational databases**.



Types of SQL Language

1. DDL – Data Definition Language

Used to define or change the **structure of database** objects.

Commands: **CREATE, ALTER, DROP, TRUNCATE**

2. DML – Data Manipulation Language

Used to **manage data** within tables

Commands: **SELECT, INSERT, UPDATE, DELETE**

3. DCL – Data Control Language

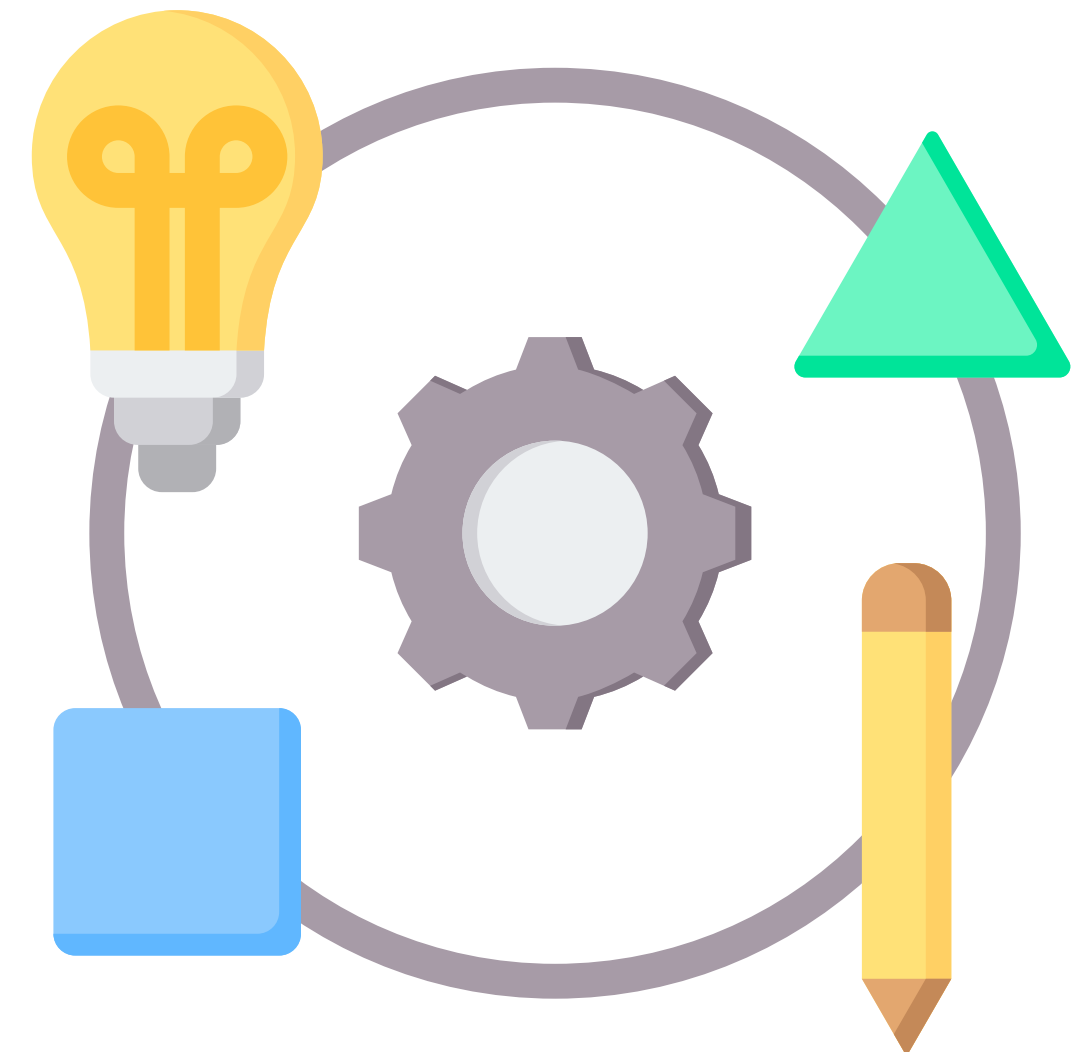
Used to **control access** and permissions

Commands: **GRANT, REVOKE**

4. TCL – Transaction Control Language

Used to **manage transactions** in a database.

Commands: **COMMIT, ROLLBACK, SAVEPOINT**



Software Created Using SQL

- Several Database Management Systems (DBMS) were built using the SQL language as their core for **querying and managing data.**

Popular Software :

- MySQL – Open-source, widely used in web applications
- Oracle Database – High-performance enterprise DBMS
- Microsoft SQL Server – For Windows-based enterprise solutions
- PostgreSQL – Advanced open-source DBMS with rich SQL support



Database Management System (DBMS)

- A Database Management System (DBMS) is a software tool that allows users to **store, access, manage, and organize data** efficiently in a database.
- The database and DBMS software together is called as Database system.



MySQL

MySQL is a popular open-source **relational database** management system (RDBMS) that uses SQL (Structured Query Language) to manage and interact with data.



Advantages

- Open-source & Free
- Fast & Reliable
- Easy to Use
- Cross-Platform Support
- Strong Community Support

Disadvantages

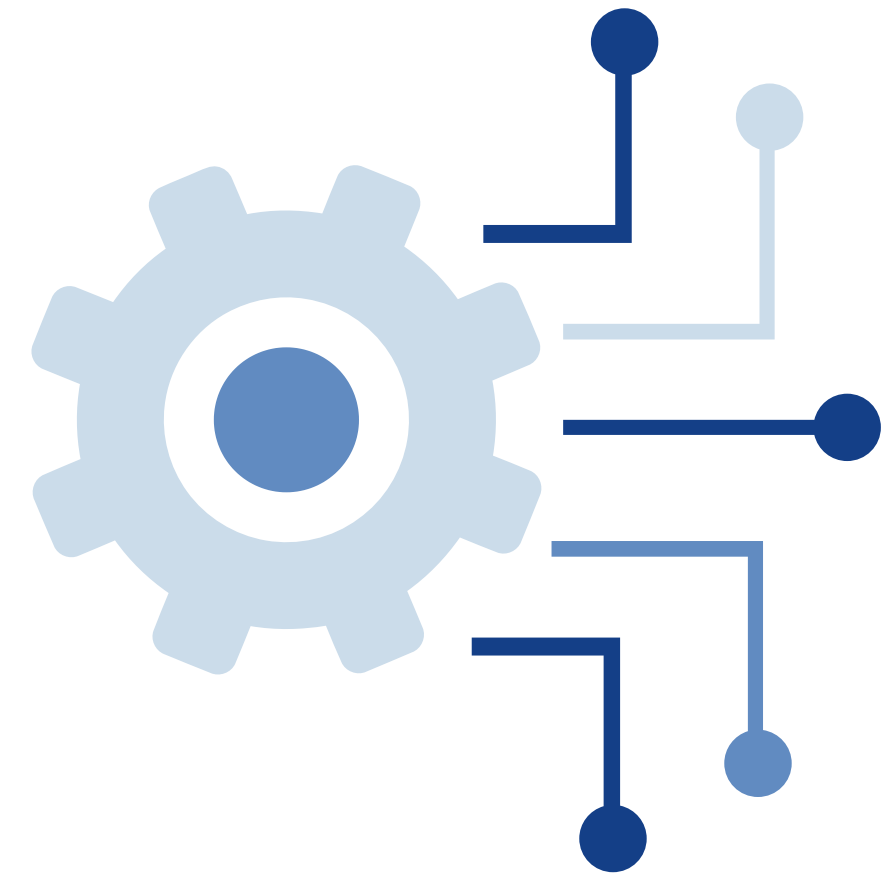
- Limited for Complex Applications
- Weaker in Handling Large Databases
- Less Support for Full SQL Standards

Client-Server Architecture in DBMS

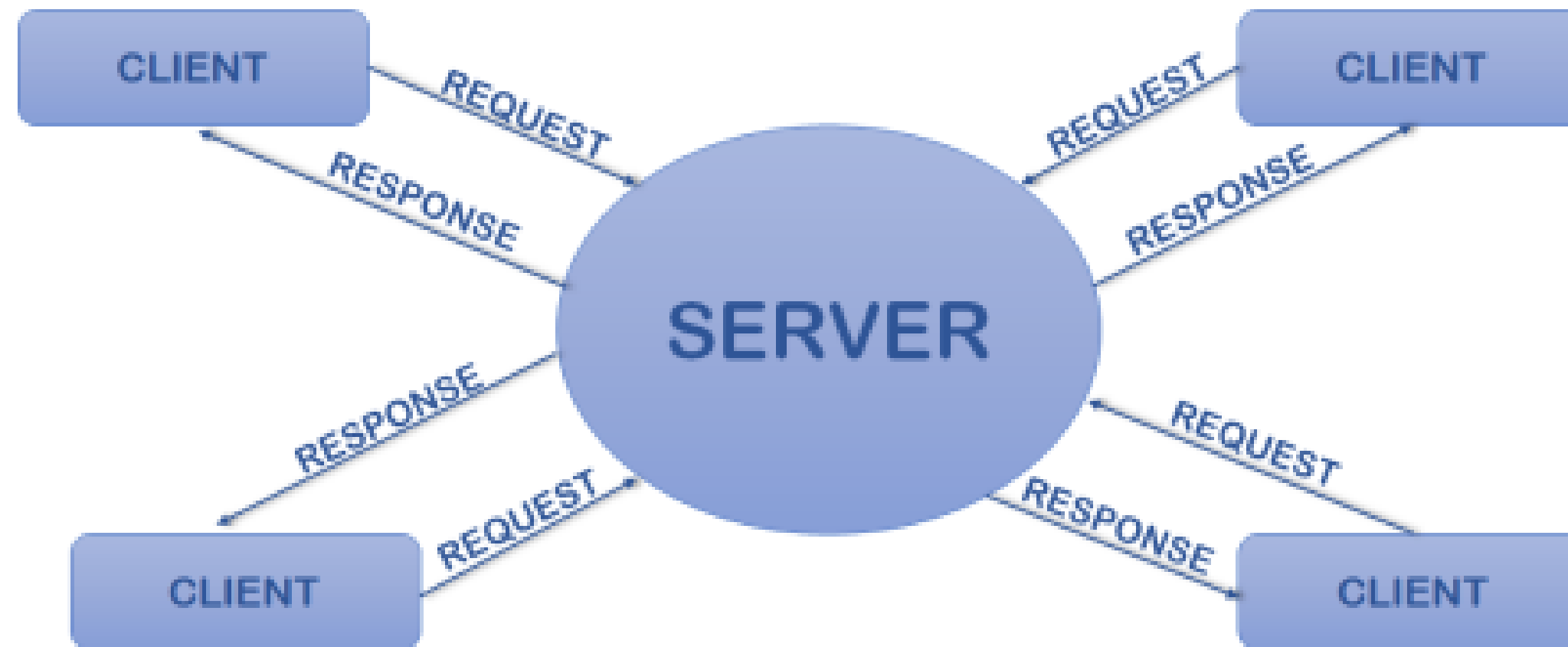
Client : User interface (e.g., desktop app, web browser, mobile app) for sending requests and displaying data.

Server : Manages and stores the database, processes client requests, and returns results.

Communication : Client and server interact over a network, with the client sending requests and the server responding.



Client and Server



MySQL Data Types

Numeric Data Types:

INT : Used to store whole numbers within a specified range

FLOAT : Used to store single-precision floating-point numbers.::

DOUBLE : Used to store double-precision floating-point numbers

DECIMAL : Used to store exact numeric values with a specified precision and scale

String Data Types:

VARCHAR : Used to store variable-length strings with a maximum length

CHAR : Used to store fixed-length strings with a specified length

TEXT : Used to store large strings of text.

Date and Time Data Types :

- **DATE** : Used to store a date (year, month, and day)
- **TIME** : Used to store a time (hour, minute, and second).
- **DATETIME** : Used to store a date and time combination.
- **TIMESTAMP** : Used to store a timestamp representing a specific point in time.

Boolean Data Type :

- **BOOLEAN or BOOL** : Used to store boolean values (true or false).

Binary Data Types:

- **BINARY** : Used to store fixed-length binary data
- **VARBINARY** : Used to store variable-length binary data.
- **BLOB** : Used to store large binary objects.